

Protecting Data Transit

1. Encrypt a sample text file (such as your Spotify playlist export) using any open-source tool like VeraCrypt or 7-Zip, then decrypt it and verify the contents match the original.

***Hint:** Use password protection and AES-256 encryption if available.*

ANS

1. Create a sample text file named playlist.txt.
 2. Open 7-Zip (or VeraCrypt).
 3. Right-click the file and select 7-Zip → Add to Archive.
 4. Choose the following settings:
 - Archive Format: 7z
 - Encryption Method: AES-256
 - Enter a Password
 5. Click OK to create the encrypted archive.
 6. Delete or move the original file (optional).
 7. Open the encrypted archive and enter the password.
 8. Extract the file.
 9. Compare the extracted file with the original.
- ❖ Observation
- The decrypted file matched the original file exactly, confirming that the encryption and decryption process was successful. AES-256 encryption protected the file from unauthorized access while it was stored or transferred.
2. Capture network traffic while logging into your Instagram or Zomato account on a web browser using Wireshark or Fiddler, and identify whether your login data is transmitted over HTTPS.

***Hint:** Look for the padlock icon in your browser and check the protocol in captured packets.*

ANS

1. Start **Wireshark**.
2. Select the active network interface.
3. Begin capturing packets.
4. Open a browser and visit **Instagram** or **Zomato**.
5. Log in to your account.
6. Stop the packet capture.
7. Apply the filter:
8. Observe the captured traffic.

❖ Observation

- The browser displayed a **padlock icon**, indicating a secure HTTPS connection.
 - The captured packets showed **TLS (Transport Layer Security)** traffic.
 - No usernames or passwords were visible in plain text because the login data was encrypted before being transmitted.
3. Use the open-source tool 'Nikto' or 'OWASP ZAP' to run a basic vulnerability scan on a local web app (such as a demo site running on localhost) and record at least 3 security issues found.

[ANS](#)

Security Issue	Risk Level	Description
Missing Security Headers	Medium	Important HTTP security headers such as Content-Security-Policy or X-Frame-Options were not configured.
Directory Browsing Enabled	Medium	Users could browse the contents of certain directories because directory listing was enabled.
Server Information Disclosure	Low	The web server revealed version information in HTTP response headers, which could help an attacker identify known vulnerabilities.

4. Perform a simulated security audit by reviewing the permissions of files in your cloud storage (Google Drive or OneDrive). List any files that are shared publicly or with 'Anyone with the link' and suggest how to restrict access for better security.

[ANS](#)

File	Sharing Status	Security Recommendation
ProjectReport.pdf	Anyone with the link	Restrict access to specific users only.
TeamPresentation.pptx	Shared with team members	No changes required.
Resume.pdf	Public	Remove public access and share only with authorized users.

❖ Recommendation

- Remove public sharing where it is not necessary.
- Share files only with specific email addresses.
- Assign **Viewer** permission instead of **Editor** whenever possible.
- Regularly review sharing settings to prevent accidental exposure of sensitive data.

5. After running a vulnerability scan, pick one reported issue (such as 'Directory Listing Enabled') and write a short action plan describing how you would fix this issue in a real cloud environment.

ANS

❖ Action Plan

1. Disable directory listing in the web server configuration (for example, Apache or Nginx).
2. Ensure that each web directory contains a default page such as **index.html** or **index.php**.
3. Review file and folder permissions to prevent unauthorized access.
4. Restart the web server to apply the configuration changes.
5. Run another vulnerability scan to verify that the issue has been resolved.

❖ Result

- Disabling directory listing helps protect sensitive files, reduces information disclosure, and strengthens the overall security of the cloud-hosted application.